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Relationship of the Phenotypic Expression of the A-I_{Milano} Apoprotein with Plasma Lipid and Lipoprotein Patterns

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Summary

The plasma lipoprotein and apolipoprotein profile of 29 adult A-I_{Milano} (A-I_M) carriers and 29 age- and sex-matched non-affected subjects of the same kindred was examined, in order to investigate linkages between the lipid and apoprotein abnormalities and the phenotypic expression of the biochemical disorder. Carriers (A-I_M^+) showed a higher prevalence of hypertriglyceridemia (12 out of 29); they also had lower plasma total cholesterol, esterified cholesterol and phospholipids, compared to non-carriers. Lipoproteins were characterized by a significant enrichment of triglycerides in low and high density fractions (LDL and HDL), and by the expected striking reduction of HDL mass and cholesterolemia. Conversely, no significant alterations of the major circulating apolipoprotein levels, except for apo A-I and apo A-II, were noted in the A-I_M^+. The increased free cholesterol/esterified cholesterol ratio in plasma (most marked in HDL), was accompanied by a significant reduction of the lecithin cholesterol acyl transferase molar activity. Several correlations pertaining to lipids, lipoproteins and apoproteins were examined: cholesterol and

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triglycerides in HDL and, more remarkably, apoprotein A-I and C-III levels in plasma were significantly correlated in the A-I_M^+.

While there was no significant prevalence of specific apo E phenotypes, plasma triglycerides and apo C-II levels were highly correlated in the carriers. The A-I_M subjects, while in the presence of severe lipoprotein risk factors, may have alternative mechanisms of cholesterol disposal, potentially responsible for the apparently low prevalence of atherosclerosis.

Key words: *A-I_{Milano} – High density lipoproteins – Hypertriglyceridemia – LCAT – Lipoprotein composition*

Introduction

Circulating low density and high density lipoproteins (LDL and HDL) are involved differently in the atherosclerotic process [1]. The association between elevated plasma levels of LDL cholesterol (LDL-Ch) and coronary heart disease (CHD) [2,3] as well as the inverse relationship between HDL cholesterol (HDL-Ch) and the risk of CHD have been suggested, based on epidemiological data [4,5].

However, the uncertainties of the antiatherogenic role of HDL have been supported by the extreme variability of CHD risk in the rare familial disorders characterized by low plasma HDL-Ch. In Tangier disease, normal HDL are absent but the risk of CHD is only slightly raised [6]; similarly, in "fish-eye" disease [7] and in familial lecithin-cholesterol acyltransferase (LCAT) deficiency [8], markedly reduced HDL levels do not appear to lead to premature cardiovascular lesions. On the other hand, the absence of the major HDL apoprotein, apo A-I [9,10], as well as familial hypoalphalipoproteinemia [11], have been associated with severe CHD. We recently reported the identification of new carriers of the A-I_{Milano} (A-I_M) apoprotein variant, all belonging to the same large kindred and characterized by a marked reduction of HDL-Ch levels [12]. The apparent lack of association between low HDL-Ch in the carriers and an increased cardiovascular risk suggested that the A-I_M variant might be an independent protective factor against CHD.

A large group of subjects with the same disorder of apolipoprotein metabolism, A-I_M carriers (A-I_M^+), was investigated in terms of their plasma lipoprotein and apoprotein profiles by comparing them to age- and sex-matched non-affected subjects of the same A-I_M kindred (A-I_M^-).

Materials and Methods

Subjects

The individual characteristics of the A-I_M carriers have been reported [12]. Plasma lipoprotein analyses were carried out in all the 29 A-I_M^+ subjects above 10 years of age. Controls, age- and sex-matched individuals (A-I_M^-) were selected by a computerized procedure between non-affected close and distant relatives from the

same kindred [12]. Detailed data of the controls and summary data from the A-I_M carriers are reported in Table 1.

Blood samples

Blood was collected on Na_2-EDTA (1 mg/ml) after an overnight fast; plasma was obtained by centrifugation at $2\,500 \times g$ and processed for lipoprotein analyses within a week. Aliquots were frozen at $-80°C$ and stored for the determination of plasma LCAT activity and the quantitation of plasma apoprotein levels.

Lipoproteins

Plasma lipoproteins were separated into density classes by sequential ultracentrifugation [13] in a type 50 titanium rotor (Beckman, Palo Alto, CA, U.S.A.). Very low density lipoproteins (VLDL) were recovered following centrifugation at $105\,000 \times g$ for 20 h. The $d = 1.006$–1.063 g/ml fraction (LDL) was prepared by raising the density to $d = 1.063$ g/ml with solid KBr and centrifuging at the same speed for 24 h. The $d = 1.063$–1.21 g/ml fraction (HDL) was isolated by centrifugation at $105\,000 \times g$ for 36 h, followed by a recentrifugation for 24 h at $d = 1.21$ g/ml. All fractions were dialyzed extensively against 0.15 M NaCl, 10 mM EDTA, pH 7.4.

Total cholesterol (TCh) and triglyceride (TG) determinations were carried out on plasma and isolated lipoproteins by enzymatic automated methods [14,15]. Free cholesterol (FCh) was determined by the same enzymatic procedure used for TCh, but omitting the esterase in the preparation of the reagent. Cholesteryl ester (CE) mass was calculated as (TCh-FCh) $\times$ 1.68. Protein analysis was according to [16] in the presence of 0.1% SDS, and phospholipids (PL) according to [17]. Aliquots of whole plasma and isolated lipoproteins were subjected to agarose gel electrophoresis [18].

Quantitative determination of plasma apoproteins

Quantitative evaluation of apoproteins was carried out by radial immunodiffusion (RID). Apo B was determined on commercial immunoplates (M-Partigen Beta Lipoproteins, Behringwerke AG, Marburg-Lahn, F.R.G.). As a standard, a reference serum of known apo B content, purchased from Behringwerke, was used. A-I, A-II, C-II and C-III were evaluated with monospecific antisera, prepared as described elsewhere [19]. The immunological determination of apo C-II and C-III was performed on the tetramethylurea (TMU) soluble fraction of native plasma. The evaluation of apo E was performed with a monospecific antiserum (kindly provided by Dr. Petar Alaupovic, Oklahoma Medical Research Foundation, Oklahoma City, OK, U.S.A.). Purified standards were prepared for the determinations and used as above described; a reference plasma was also included in each run.

Apoprotein composition

The composition of isolated lipoprotein fractions was determined by SDS electrophoresis and analytical isoelectric focusing (IEF). Lipoproteins were dialyzed against 0.01% EDTA, pH 7.4 and lyophilized; apoproteins were prepared by chloroform/methanol (2 : 1, v/v) delipidation [20].

Apoproteins for SDS electrophoresis were incubated for 90 min at 37°C in a 0.125 M Tris HCl buffer (pH 6.8) containing SDS (0.3%) and 2-mercaptoethanol (1%). Electrophoresis was carried out on 14% acrylamide slab gels (140 × 120 × 1.5 mm) by the method of Laemmli [21]. IEF was performed on 7.5% acrylamide, 8 M urea slab gels containing 2% Ampholine (LKB, Bromma, Sweden); the pH range 4–6 was used for HDL apoproteins and for the C-peptides of VLDL, whereas a pH range 5–7 was employed for the apo E phenotyping. A semi-quantitative evaluation was carried out by measuring the relative intensity of the apoprotein bands [22].

The apo B content of VLDL and LDL was determined by a TMU selective precipitation method [23]. The apo A-I content of HDL was measured by radial immunodiffusion as described for plasma.

LCAT activity

LCAT activity was determined according to [24], with [4-^{14}C]-cholesterol (0.12 μCi, 58.4 mCi/mmole) as the labelled substrate in 100 μl of plasma. After a 4-h equilibration at 37°C, 2-mercaptoethanol was added (4 mM final) and incubation continued for 2 h. The reaction was stopped by chloroform/methanol (2 : 1, v/v). The organic phase was chromatographed on Silica gel plates (Merck, Bracco, Italy) with hexane/ethylether/acetic acid (70 : 30 : 1). The conversion of [4-^{14}C]-cholesterol was determined by scanning the plates with a radiochromatoscanner (Berchtold, F.R.G.) and by measuring the areas of the radioactive peaks migrating with the Rf of standard cholesterol and of cholesterol oleate. The LCAT activity was calculated assuming complete equilibration of the labelled cholesterol with the endogenous free cholesterol [24].

Statistics

Statistical analyses were carried out by the Student *t*-test and *r* correlation coefficients [25].

Results

The two series of individuals, A-I_M^+ and A-I_M^- proved to be quite comparable, apart from the lipid findings (Table 1). A-I_M^- subjects had a higher mean body mass index (BMI) and higher systolic and diastolic blood pressures. These two latter findings, not significantly different between the two groups, are possibly related to a higher prevalence of hypertension in the A-I_M^-. On the whole, there were 7 clinically hypertensive subjects among the controls vs 4 among the A-I_M carriers.

Plasma lipid and apoprotein findings

Plasma lipid and lipoprotein analyses were carried out in 12 female and 17 male subjects with or without the A-I_M variant (Table 1). Twelve of the 29 A-I_M^+ were identified as hypertriglyceridemic (HTG) on the basis of the plasma lipoprotein electrophoretic profile and of a plasma TG value above the 90th percentile for the respective age and sex class, recorded from the total population of Limone sul Garda [26]. By contrast, only 4 of the age- and sex-matched A-I_M^- showed a TG elevation; one was hypercholesterolemic (type II-A).

TABLE 1

AGE, SEX, BLOOD PRESSURE AND BIOCHEMICAL CHARACTERISTICS OF THE AGE- AND SEX-MATCHED CONTROLS ($A\text{-}I_M^-$) OF THE $A\text{-}I_M$ CARRIERS ($A\text{-}I_M^+$) [12]

Kindred position	Age (yr)	Sex	Plasma cholesterol (mg/dl)	Plasma triglycerides (mg/dl)	BMI (kg/m^2)	Blood pressure (mm Hg)	
						Systolic	Diastolic
V-14	89	M	255	127	21.0	125	85
V-90	75	M	198	89	25.4	155	80
VI-18	56	M	190	119	36.3	160	80
VI-20	57	F	222	132	29.0	180	105
VI-60	46	M	228	116	27.3	140	95
VI-125	60	M	245	105	32.9	210	115
VI-129	43	M	268	211	28.4	140	95
VI-133	55	M	273	225	31.4	160	100
VI-136	54	F	330	126	30.5	160	90
VI-180	38	F	190	58	17.2	110	80
VI-183	34	M	208	109	21.4	130	85
VI-201	60	F	285	196	35.0	155	95
VI-202	56	M	200	72	25.5	120	80
VII-42	36	F	219	87	22.7	135	90
VII-44	32	F	232	73	20.4	130	90
VII-45	29	M	178	60	30.8	165	115
VII-47	18	F	140	62	22.9	125	80
VII-138	20	F	130	87	21.8	145	90
VII-139	15	F	152	58	18.0	130	80
VII-140	13	F	191	127	27.4	130	80
VII-141	11	M	200	64	21.0	110	75
VII-150	28	M	347	234	27.6	150	100
VII-152	19	F	198	78	22.3	130	80
VII-172	10	M	176	54	14.8	100	75
VII-193	38	M	266	150	24.5	150	100
VII-203	32	F	223	88	25.4	145	90
VII-204	28	M	187	89	26.8	145	100
VII-205	16	M	162	67	25.6	135	90
VIII-25	11	M	170	48	15.2	100	70
$A\text{-}I_M^-$							
$x \pm SD$	37.2 ± 20.5	12F +17M	216.0 ± 52.0	107.3 ± 52.2	25.1 ± 5.5	140.3 ± 23.5	89.1 ± 11.3
range	11–89		130–347	48–234	14.8–36.3	100–210	70–115
$A\text{-}I_M^+$							
$x \pm SD$	37.9 ± 19.7	12F +17M	181.5 ± 45.2	187.9 ± 3.0	23.9 ± 3.0	130.3 ± 23.8	81.7 ± 12.3
range	13–81		125–298	61–336	18.7–30.9	90–200	60–120

Due to the high number of HTG subjects in the $A\text{-}I_M^+$ group, the mean plasma TG level was significantly higher in the total group of $A\text{-}I_M^+$ compared to $A\text{-}I_M^-$ (Table 2); no difference in plasma TG levels could be detected between the $A\text{-}I_M^+$

TABLE 2
PLASMA LIPID LEVELS IN THE A-I_{Milano} CARRIERS (A-I_M^+) AND AGE- AND SEX-MATCHED CONTROLS (A-I_M^-), BOTH NORMOLIPIDEMIC (NL) AND HYPERTRIGLYCERIDEMIC (HTG)

Values are expressed as ±SD.

		Total cholesterol (mg/dl)	Free cholesterol (mg/dl)	Esterified cholesterol (mg/dl)	Triglycerides (mg/dl)	Phospholipids (mg/dl)
A-I_M^+	29	177.3 ± 43.2 °	52.6 ± 14.7	124.7 ± 29.8	184.5 ± 78.5 •	194.1 ± 52.5 °
A-I_M^+ HTG	12	211.2 ± 35.1 •,☆	65.9 ± 9.2 ☆	145.3 ± 27.5 •,☆	262.8 ± 43.8 ☆	226.5 ± 49.8 •,*
A-I_M^+ NL	17	153.4 ± 30.7 •	43.1 ± 9.7	110.1 ± 22.2 •	129.2 ± 40.2	171.2 ± 42.2 °
A-I_M^-	29	209.7 ± 59.5	51.1 ± 14.4	158.7 ± 45.9	126.5 ± 70.1	223.3 ± 52.7
A-I_M^- HTG	4	299.8 ± 46.2 ☆	70.5 ± 6.7 ☆	229.3 ± 39.6 ☆	265.5 ± 57.9 ☆	335.5 ± 50.2 ☆
A-I_M^- NL	25	195.2 ± 47.8	48.0 ± 12.7	147.3 ± 36.0	104.2 ± 39.8	209.3 ± 40.3

° $P < 0.05$, • $P < 0.005$ vs A-I_M^- in the same subdivision.
* $P < 0.05$, ☆ $P < 0.001$ vs NL in the same group.

and A-I_M^- with HTG, whereas normolipidemic (NL) A-I_M^+ showed slightly higher TG levels compared to NL A-IM_M^-.

Plasma TCh was significantly lower in all the A-I_M^+ compared to the A-I_M^-; higher TCh levels were detected in both HTG groups, compared to NL (Table 2). Despite the significant reduction of TCh in the A-I_M^+, the FCh levels in both groups were similar, and plasma PL constantly higher in A-I_M^- compared to A-I_M^+.

Plasma apoprotein findings are reported in Table 3. A major problem in the determination of plasma and lipoprotein-associated A-I may arise from the possible differences in the immunological properties between normal A-I, used for the antiserum production, and the various molecular forms of A-I_M. When tested

TABLE 3
PLASMA APOLIPOPROTEIN LEVELS IN THE A-I_{Milano} CARRIERS (A-I_M^+) AND IN AGE- AND SEX-MATCHED CONTROLS (A-I_M^-), BOTH NORMOLIPIDEMIC (NL) AND HYPERTRIGLYCERIDEMIC (HTG)

Values are mg/dl and are expressed as $x \pm SD$.

		A-I	A-II	B	C-II	C-III	E
A-I_M^+	29	73.8 ± 19.8 •	24.1 ± 4.8 •	86.4 ± 24.7	4.79 ± 0.65	12.62 ± 1.98	6.21 ± 1.90
A-I_M^+ HTG	12	73.4 ± 24.8 •	23.5 ± 4.9 °	103.8 ± 23.7 *	5.00 ± 0.54	13.41 ± 2.90	7.38 ± 2.22
A-I_M^+ NL	17	74.1 ± 16.2 •	24.6 ± 4.8 •	74.1 ± 17.3	4.68 ± 0.68	12.25 ± 1.24	5.63 ± 1.48
A-I_M^-	29	128.6 ± 17.9	38.2 ± 7.5	80.9 ± 23.6	4.71 ± 0.55	13.04 ± 2.44	6.59 ± 1.99
A-I_M^- HTG	4	133.0 ± 28.2	42.5 ± 9.9	118.8 ± 11.3 *	4.63 ± 0.42	13.37 ± 1.59	8.30 ± 1.51
A-I_M^- NL	25	127.9 ± 16.4	37.6 ± 7.0	74.8 ± 18.9	4.72 ± 0.57	13.00 ± 2.56	6.32 ± 1.96

° $P < 0.05$, • $P < 0.001$ vs A-I_M^- in the same subdivision.
* $P < 0.001$ vs NL in the same group.

TABLE 4

PLASMA LIPOPROTEIN LEVELS, EXPRESSED AS MASS AND AS CHOLESTEROL CONCENTRATION, IN THE A-I_{Milano} CARRIERS (A-I^+_M) AND IN AGE-AND SEX-MATCHED CONTROLS (A-I^-_M), BOTH NORMOLIPIDEMIC (NL) AND HYPERTRIGLYCERIDEMIC (HTG)

Values are reported as mg/dl and expressed as $x \pm SD$.

		Very low density lipoproteins		Low density lipoproteins		High density lipoproteins	
		Mass	Cholesterol	Mass	Cholesterol	Mass	Cholesterol
A-I^+_M	29	192.6 ± 101.2 °	321.9 ± 13.5	483.6 ± 126.4	136.9 ± 36.4	176.1 ± 79.2 •	17.8 ± 8.1 •
A-I^+_M HTG	12	289.2 ± 71.3 ☆	34.3 ± 12.0 °,☆	586.6 ± 107.3 ☆	161.8 ± 35.0 ☆	163.3 ± 88.2 •	14.9 ± 7.4 •
A-I^+_M NL	17	124.4 ± 49.7	13.2 ± 4.9	410.8 ± 80.4	119.4 ± 26.3	185.2 ± 73.6 •	19.9 ± 8.1 •
A-I^-_M	29	134.6 ± 97.7	16.0 ± 13.8	449.3 ± 171.7	140.9 ± 49.4	371.4 ± 78.7	54.2 ± 11.7
A-I^-_M HTG	4	332.0 ± 55.6 ☆	45.2 ± 6.5 ☆	658.8 ± 80.5 ☆	196.8 ± 35.1 *	425.5 ± 116.3	57.8 ± 14.8
A-I^-_M NL	25	103.0 ± 56.6	11.4 ± 7.2	415.8 ± 158.4	132.0 ± 45.7	362.8 ± 70.5	52.5 ± 11.3

° $P < 0.05$, • $P < 0.001$ vs A-I^-_M in the same subdivision.

* $P < 0.005$, ☆ $P < 0.001$ vs NL in the same group.

against the apo A-I antiserum by immunodiffusion, the immunochemical identity of A-I_M, A-I_M–A-I_M and A-I_M–A-II with normal A-I could be demonstrated [27]. Furthermore, standard curves prepared with purified A-I and A-I_M show similar slopes (A-I = 0.106; A-I_M = 0.105) and intercepts (A-I = 0.98; A-I_M = 1.00) when tested by RID. To standardize the technique for the presence of A-I_M aggregates, 5 HDL samples from different A-I_M carriers, with variable amounts of A-I_M–A-I_M and A-I_M–A-II (as assessed by SDS gel electrophoresis), were analyzed by RID both in native and reduced conditions, giving identical readings. The presented results refer, therefore, to the total amount of A-I_M present in the carrier plasma and/or lipoproteins.

Highly significant differences between A-I_M^+ and A-I_M^- were found in the A apoprotein levels, apo A-I and A-II being markedly lower in the A-I_M^+; the A-I/A-II weight ratio in plasma was significantly reduced in the A-I_M^+ (2.99 ± 0.54) compared to the A-I_M^- (3.45 ± 0.66; $P < 0.01$). The plasma concentration of the other tested apoproteins, i.e. apo B, C-II, C-III and E, did not show appreciable differences. As a consequence, the A-I/B apoprotein ratio, reported to be a good discriminating index for atherosclerosis risk [28], was significantly lower for the A-I_M^+ (0.93 ± 0.36), compared to the A-I_M^- (1.70 ± 0.47, $P < 0.001$).

Lipoprotein levels and composition

Despite the increased prevalence of hypertriglyceridemia in the A-I_M^+ group, the plasma levels of VLDL were only slightly higher, compared to the A-I_M^- (Table 4). No difference could be detected in the LDL concentrations, although the LDL-Ch to plasma apo B ratio was slightly lower in the A-I_M^+ (1.61 ± 0.28 vs 1.75 ± 0.40 for A-I_M^-). Higher LDL mass and Ch levels were determined in hypertriglyceridemic subjects, both A-I_M^+ and A-I_M^-. HDL were markedly reduced in the A-I_M^+, both in terms of total mass (calculated from the sum of individual components) and of lipoprotein associated cholesterol (Table 4). HDL-Ch in the A-I_M subjects is about one third and total HDL concentration about one half of those detected in A-I_M^-. The HDL-Ch to plasma A-I ratio is significantly lower in the A-I_M^+ (0.22 ± 0.06 vs 0.36 ± 0.09, $P < 0.001$).

The weight percentages of protein and lipids in the VLDL, LDL and HDL fractions, isolated from A-I_M^+ and A-I_M^-, are reported in Table 5. VLDL composition is similar in the two groups, the protein content in this fraction being slightly higher, compared to the 8–9% value reported by others [29], for normal subjects. LDL composition in the A-I_M^+ differed from A-I_M^- in the lipid constituents of the lipoprotein core (Table 5). In the A-I_M^+, the percentage of CE was about 4% lower, with a higher content of TG. This finding is probably not due to the higher prevalence of HTG subjects, because the LDL-TG content was similarly elevated in the NL A-I_M^+ subjects, compared to the corresponding NL A-I_M^- ($12.1 \pm 2.4\%$ vs $8.8 \pm 2.4\%$, $P < 0.001$). The estimates for protein and lipid content of LDL in A-I_M^- are within the range of values reported for normal subjects [29].

HDL of markedly abnormal composition were isolated from the A-I_M^+ (Table 5). The anomalous HDL are TG-, PL- and protein-enriched, with a marked reduction of the CE content. The same differences could be detected when HTG or NL

TABLE 5

PLASMA LIPOPROTEIN COMPOSITION IN THE 29 A-I_{Milano} CARRIERS (A-I^+_M) AND IN AGE- AND SEX-MATCHED CONTROLS (A-I^-_M)

Values are reported as percentages for each group of 29 subjects and are expressed as $x \pm SD$.

	Free cholesterol	Cholesteryl esters	Triglycerides	Phospholipids	Protein
Very low density lipoproteins					
A-I^+_M	4.7 ± 0.7	11.3 ± 3.0	55.0 ± 6.3	18.1 ± 3.8 °	11.4 ± 3.0
A-I^-_M	4.3 ± 0.9	11.4 ± 3.0	55.9 ± 6.0	16.1 ± 3.2	12.3 ± 3.2
Low density lipoproteins					
A-I^+_M	8.1 ± 1.2	34.2 ± 4.4 °	13.9 ± 3.1 •	23.2 ± 4.7	21.1 ± 2.2
A-I^-_M	7.6 ± 1.7	38.0 ± 5.1	9.1 ± 2.7	23.5 ± 3.0	20.9 ± 2.4
High density lipoproteins					
A-I^+_M	2.7 ± 0.6	12.4 ± 3.0 •	6.4 ± 2.2 •	26.5 ± 3.6 °	52.2 ± 4.5 °
A-I^-_M	2.6 ± 0.5	19.6 ± 2.6	3.0 ± 1.1	24.6 ± 2.7	49.7 ± 3.3

° $P < 0.05$, • $P < 0.001$ significantly different from A-I^-_M.

carriers were compared to A-I^-_M in the same metabolic condition (*HTG*: CE = 11.2 ± 1.4% for A-I^+_M vs 19.0 ± 1.0% for A-I^-_M, $P < 0.001$; TG = 7.6 ± 2.2% vs 4.1 ± 1.2%, $P < 0.005$; *NL*: CE = 13.3 ± 3.6% vs 19.7 ± 2.8%, $P < 0.001$; TG = 5.5 ± 1.8% vs 2.8 ± 1.0%, $P < 0.001$). The molar ratios between the various lipid classes in VLDL, LDL and HDL from A-I^+_M and A-I^-_M are consistent for each lipoprotein fraction showing minor differences between the 2 groups. The CE/TG molar ratio is lower in the LDL and HDL isolated from the A-I^+_M (LDL: 3.44 ± 1.11 vs 6.06 ± 2.16, $P < 0.001$; HDL: 2.97 ± 1.49 vs 9.82 ± 4.00, $P < 0.001$); the calculated surface to core ratios are similar for VLDL and LDL, but HDL from the A-I^+_M subjects show a marked enrichment in surface components (A-I^+_M = 2.38 ± 0.63; A-I^-_M = 1.97 ± 0.35, $P < 0.005$).

The apoprotein composition of VLDL was similar in all subjects. No difference could be detected in the apo E phenotype distribution in the two groups. None of the subjects showed a E2/2, E4/4 or E2/4 phenotype. The frequency of the E3/3, E2/3 and E3/4 phenotypes was: 62.0%, 24.1% and 13.8% for A-I^+_M and 58.6%, 27.6% and 13.8% for A-I^-_M, i.e. close to expected percentages in the general population [30]. No difference was detected in the distribution of apo E phenotypes between NL and HTG subjects, both in the A-I^+_M and A-I^-_M groups. In LDL, apo B constituted 95.2 ± 2.0% of total proteins in the A-I^+_M and 93.4 ± 4.6% in the A-I^-_M.

Important differences between A-I^+_M and A-I^-_M were detected in the apoprotein composition of HDL. The A-I/A-II ratio, as measured by semi-quantitative SDS electrophoresis and IEF of reduced HDL apoproteins, was significantly lower in the A-I^+_M (2.72 ± 0.43) compared to the A-I^-_M (3.35 ± 0.54, $P < 0.001$).

The A-I content of HDL was also determined by radial immunodiffusion, as reported for plasma. The recovery of plasma A-I in the HDL fraction was 86.1 ±

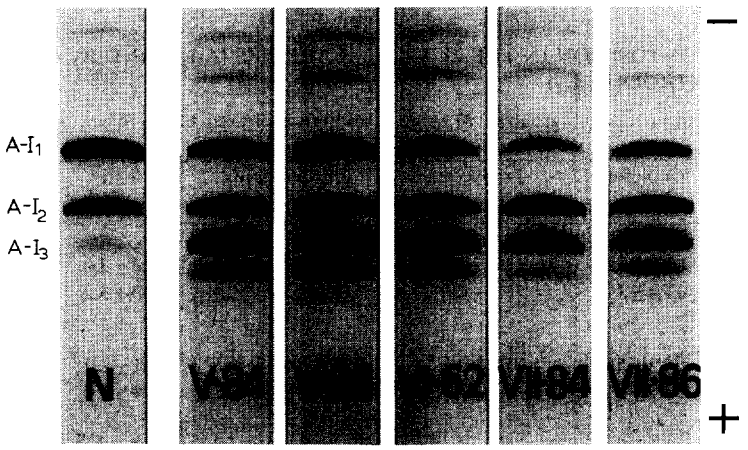


Fig. 1. Isoelectric focusing pattern (pH 4–6) of mercaptoethanol reduced HDL apoproteins isolated from a normal subject (N) and from 5 A-I_M carriers, identified by the kindred position [12]. A striking variability in the relative content of isoprotein A-I_1, previously identified as normal A-I in the affected subjects [27], is apparent in the different A-I_M carriers.

17.8% for A-I_M^- and 82.1 ± 15.3% for A-I_M^+ (*P*: ns). By analytical IEF, the relative abundance of isoprotein A-I-1, the major isoprotein of apo A-I in normal plasma, corresponding to the normal A-I in the A-I_M^+ [31], ranged between 9.3% and 35.3% of total apo A-I (Fig. 1).

Correlations between plasma lipoprotein and apoprotein levels

In an attempt to disclose structural or dynamic associations between lipids, lipoproteins and different apoproteins, numerous correlations were calculated, referring to the complete samples of A-I_M^+ and A-I_M^- subjects. Correlations between plasma lipids and lipoproteins were, as expectable (Table 6) highly significant, both in the case of A-I_M^+ and A-I_M^-, for parameters related to plasma cholesterol transport (TCh vs VLDL-Ch and LDL-Ch, statistically significant in both groups); similarly, significant correlations were found for whole plasma TG levels (with TCh and also with VLDL-Ch and LDL-Ch).

Some differences between the 2 phenotypes were, however, striking. In particular, the A-I_M^+ showed a highly significant correlation between the TG and the Ch contents in HDL, in contrast to the A-I_M^-. Moreover, carriers showed a significantly negative correlation between VLDL lipids (both Ch and TG) and HDL-Ch, which did not achieve close to statistical significance in the A-I_M^-. HDL-Ch levels were also negatively correlated with triglyceridemia in the A-I_M^+, thus confirming [22]. In contrast, there was no correlation in the A-I_M^+ between triglyceridemia and the TG content of HDL.

Correlations pertaining to plasma apoproteins, lipids and BMI, followed the expected pattern, with significant correlations between apo A-I and A-II, TG and apo E, TG and BMI [32,33]. On the other hand, some differences between the 2

TABLE 6

CORRELATION COEFFICIENTS BETWEEN TOTAL CHOLESTEROL (TCh) AND TRIGLYCERIDES (TG) IN PLASMA AND IN THE DIFFERENT LIPOPROTEIN FRACTIONS IN THE 29 A-I_{Milano} CARRIERS (A-I_M^+) AND IN THE AGE- AND SEX-MATCHED CONTROLS (A-I_M^-)

	VLDL-TG		LDL-TCh		LDL-TG		HDL-TCh		HDL-TG		Plasma-TCh		Plasma-TG	
	A-I_M^+	A-I_M^-	A-I_M^+	A-I_M^-	A-I_M^+	A-I_M^-	A-I_M^+	A-I_M^-	A-I_M^+	A-I_M^-	A-I_M^+	A-I_M^-	A-I_M^+	A-I_M^-
VLDL-TCh	0.89 *	0.96 *	0.55 ☆	0.61 *	0.80 *	0.78 *	−0.40 *	−0.03	−0.09	0.58 *	0.69 *	0.73 *	0.90 *	0.95 *
VLDL-TG			0.55 ☆	0.58 *	0.72 *	0.78 *	−0.37 *	−0.14	0.19	0.49 ☆	0.64 *	0.68 *	0.96 *	0.97 *
LDL-TCh					0.72 *	0.63 *	−0.15	−0.01	0.11	0.28	0.96 *	0.96 *	0.64 *	0.59 *
LDL-TG							−0.30	−0.03	−0.10	0.64 *	0.79 *	0.55 *	0.72 *	0.88 *
HDL-TCh									0.63 *	0.20	−0.07	0.18	−0.35 *	−0.10
HDL-TG											0.24	0.38 *	0.25	0.61 *
Plasma-TCh													0.74 *	0.71 *

* $P < 0.05$, ☆ $P < 0.01$, * $P < 0.001$.

TABLE 7

PLASMA MOLAR LECITHIN CHOLESTEROL ACYL TRANSFERASE (LCAT) ACTIVITY AND FREE CHOLESTEROL/ESTERIFIED CHOLESTEROL (FCh/ECh) RATIOS IN PLASMA AND IN THE DIFFERENT LIPOPROTEINS IN A-I_{Milano} CARRIERS (A-I_M^+) AND APPROPRIATE CONTROLS (A-I_M^-), BOTH HYPERTRIGLYCERIDEMIC (HTG) AND NORMOLIPIDEMIC (NL)

Data are expressed as $x \pm SD$.

		LCAT [a] (molar)	Plasma FCh/ECh	VLDL FCh/ECh	LDL FCh/ECh	HDL FCh/ECh
A-I_M^+	29	13.41 ± 4.30 •	0.42 ± 0.07 ⊙	0.74 ± 0.21	0.40 ± 0.07 •	0.39 ± 0.10 ⊙
A-I_M^+ HTG	12	15.56 ± 5.14 °	0.46 ± 0.07 ⊙	0.76 ± 0.22	0.43 ± 0.09 ⊙	0.41 ± 0.09 ⊙
A-I_M^+ NL	17	12.21 ± 3.34 •	0.39 ± 0.05 ⊙	0.72 ± 0.21	0.38 ± 0.05	0.38 ± 0.11 ⊙
A-I_M^-	29	18.15 ± 6.50	0.32 ± 0.04	0.67 ± 0.18	0.35 ± 0.05	0.22 ± 0.04
A-I_M^- HTG	4	25.43 ± 5.59	0.31 ± 0.02	0.57 ± 0.10	0.30 ± 0.03	0.21 ± 0.04
A-I_M^- NL	25	16.94 ± 5.89	0.33 ± 0.04	0.69 ± 0.18	0.35 ± 0.05	0.22 ± 0.04

° $P < 0.01$, • $P < 0.05$, ⊙ $P < 0.001$ vs A-I_M^- in the same subdivision.

[a] nmoles cholesteryl esters/ml/h.

groups of subjects were of special interest, e.g., the highly significant correlation, detected in the A-I_M^+, between plasma C-II and C-III levels (A-I_M^+, $r = 0.519$, $P < 0.01$; A-I_M^-, $r = 0.030$, P: n.s.), and between triglyceridemia and apo C-II levels (A-I_M^+, $r = 0.391$, $P < 0.05$; A-I_M^-, $r = 0.033$, P: n.s.). Moreover, only the A-I_M^+ showed a significant correlation between A-I and C-III (A-I_M^+, $r = 0.414$, $P < 0.05$; A-I_M^-, $r = -0.216$, P: n.s.).

LCAT activity

Despite the significant hypocholesterolemia detected in the A-I_M^+, plasma FCh levels were similar in the two groups (Table 2), so that the FCh/ECh ratio in plasma was about 30% higher in the A-I_M^+ subjects compared to the A-I_M^- (Table 7). A significantly higher FCh/ECh ratio was also noted in both LDL and HDL, isolated from the carriers.

The LCAT activity, was about 25% lower in the A-I_M^+, compared to A-I_M^- (Table 7), the fractional esterification rate being $2.16 \pm 0.54\%$ vs $2.73 \pm 0.59\%$ ($P < 0.001$). In the A-I_M^+ LCAT activity was highly significantly correlated with the HDL mass (A-I_M^+, $r = 0.609$, $P < 0.001$; A-I_M^-, $r = 0.203$, P: ns), but not with plasma TG (A-I_M^+, $r = 0.327$, P: ns; A-I_M^-, $r = 0.631$, $P < 0.001$).

Discussion

These findings relate to the largest described series of carriers of a human apolipoprotein variant, and confirm the general observation that the A-I_M^+ subjects show most of the traditional lipoprotein "risk indexes" for cardiovascular disease. In particular, they have reduced HDL-Ch levels, HDL-Ch/LDL-Ch, A-I/B, A-I/A-II and LDL-Ch/B ratios [28,34]. The low A-I/A-II ratio is consistent with the

previously described marked reduction of the HDL_2 subfraction [35]. In addition, many of the carriers show significant hypertriglyceridemia. In spite of the presence of these metabolic risk indices, the $A\text{-}I_M^+$ subjects do not show a particularly elevated prevalence of cardiovascular events; they may possibly be somewhat protected from the development of arterial lesions [12].

In a general evaluation of the apparent reduction of the atherogenic risk in $A\text{-}I_M^+$ subjects, many specific alterations seem to be cooperative. The presence of HDL particles of very high density, in the almost complete absence of HDL_2 may be a promoting factor for cholesterol efflux from tissues and also for an increased LDL receptor activity [36,37]. In physiological conditions, e.g. vegetarians, associated with low cardiovascular morbidity, a significant reduction of the A-I/A-II ratio is observed [38]. Since, in previous clinical observations, it could be shown that no significant accumulation of lipids and/or apoproteins is detectable in the intestine of the $A\text{-}I_M$ carriers, it may be assumed that this tissue secretes HDL particles normally, most likely of a high density, with increased protein (apo A-I) and PL content [39].

The described reduction of LCAT activity contrasts with the initial observation on the original proband family, in which LCAT activity was determined by a non-isotopic technique [40]. A reduced enzyme activity is consistent with the observed higher FCh/ECh ratio in the plasma of carriers. The inability of the $A\text{-}I_M$ dimers and complexes with apo A-II to activate LCAT in an *in vitro* system has been recently reported [41]. However, the monomeric $A\text{-}I_M$ and the normal A-I present in the carriers are largely sufficient for LCAT activation [42], and other apolipoproteins are known to act as cofactors for the esterification reaction [43]. A normal cholesterol esterification has been detected in patients with the apo A-I deficiency [9]. The reduced LCAT activity in the $A\text{-}I_M$ carriers can thus derive from a low plasma enzyme level and/or from the inability of the anomalous HDL particles to efficiently act as a substrate. This latter hypothesis is supported by the highly significant correlation between LCAT activity and HDL mass in the carriers, but not in the controls. As a consequence [44], small, dense HDL_3 particles can be detected in the $A\text{-}I_M$ subjects [35].

These small HDL_3 are rapidly catabolized, without conversion to HDL_2, resulting in an accelerated turnover of the variant apoprotein [45], as well as an efficient removal of cholesterol from tissues, not leading to thesaurismosis or atherogenesis. A likely consequence of this deranged mechanism may be hypertriglyceridemia, as well as the obvious reduction of HDL levels, which is variable in the different subjects and related to the amount of the mutant apoprotein in HDL [22].

The hypertriglyceridemia is characterized by a slight elevation of VLDL (of normal composition) and LDL (with an increased TG content); HDL are reduced and TG enriched. It is, therefore, apparently similar to those detected in Tangier patients [19] and in subjects with fish-eye disease [7], probably not related to the presence of the variant. The reduction of the HDL pool, common to all of these familial disorders, is possibly responsible, through a derangement of lipid (particularly TG and CE) exchange between HDL and lipoproteins of lower densities [46], of the hypertriglyceridemia in the affected subjects. Furthermore, a genetic transmis-

sion of a metabolic disorder leading to elevated triglyceridemia can be excluded by the highly significant difference in the frequency of hypertriglyceridemia between the A-I_M carriers and non-carriers from the same kindred [12].

Examination of the apo E isoform pattern shows that neither among the A-I_M^+, nor among A-I_M^- subjects was there any clustering of a specific phenotype. Of particular interest was the significant association between levels of apo A-I and apo C-III in the A-I_M^+; this did not occur among the A-I_M^-. A chromosomal linkage between apo A-I and C-III has recently been described [47], occurring on chromosome 11 [48]. Gene sequences codifying for the two apoproteins are located on neighbouring tracts of the chromosome, with opposite direction [47]. A chromosomal insert is linked to the phenotypic expression of the apo A-I/C-III deficiency [49]. In view of the lack of correlation between apo A-I and C-III levels in A-I_M^-, it would appear that the phenotypic expression of the gene is differently regulated for the two sequences in normal subjects. In contrast, there may be a common regulatory mechanism for expression in the A-I_M^+. The restriction pattern of chromosomal DNA in the A-I_M^+, compared to A-I_M^-, is currently being investigated.

The low prevalence of atherosclerotic complications in the A-I_M carriers, in spite of concomitant severe lipoprotein risk factors, argues for the presence of alternative protective mechanisms, for example the presence of HDL particles of high density, the apparently reduced transfer of CE to lower density lipoproteins and the higher FCh content in HDL. The reported findings have attempted to provide detailed information on this rare metabolic disorder, characterized by a dominant transmission, where several pathophysiological mechanisms seem to operate towards a reduced risk of the development of arterial disease.

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